PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE

Department of Computer Engineering

LP-III (PR) – Student Instructions and Software Requirements

Programme: B.E. Computer Engineering

Course: LP-III (PR)

Practical Examination Dates: 02 October 2026 to 06 October 2026

Examination Venues:

- A1-303 and A1-314: 02 October 2026 to 05 October 2026
- A1-204 and A1-216: 03 October 2026 to 06 October 2026

1. General Instructions for Students

1. Each practical examination assignment will comprise any two groups selected from the following three groups: Group A (Design and Analysis of Algorithms), Group B (Machine Learning), and Group C (Blockchain Technology).
2. Prepare all five problem statements in each relevant group, as the specific combination assigned during the examination may vary.
3. Familiarize yourself with the required development environments, programming languages, libraries, compilers, and extensions listed in this document.
4. Practise writing, executing, debugging, and explaining your programs. Be prepared to demonstrate the output and explain the approach used in your solution.
5. For Machine Learning assignments, make sure you understand the required preprocessing, model-building or algorithm steps, and evaluation measures mentioned in the assigned problem statement.
6. For Blockchain Technology assignments, review smart-contract functions, compilation, deployment, and interaction with the contract. Follow the examination instructions regarding the permitted deployment environment.
7. Follow the instructions of the examiners and laboratory staff during the practical examination.

2. Software Requirements

Group / Practical Area	Development Environment	Required Software / Libraries
Group A (A1–A5): Design and Analysis of Algorithms	Eclipse IDE (Java); Visual Studio Code (C++)	Java JDK 25 and javac; GCC
Group B (B1–B5): Machine Learning	Visual Studio Code with Python 3.12.x	NumPy, pandas, Matplotlib, Seaborn, scikit-learn, SciPy, TensorFlow, NLTK and required NLTK corpora
Group C (C1–C2): Blockchain Technology	Remix IDE Desktop; MetaMask/browser wallet where required	Solidity compiler (solc) version 0.8.38; test-network access and wallet/network configuration where required

3. Practice Assignment Areas

You should prepare the following problem statements in the groups applicable to your assigned combination.

Group A: Design and Analysis of Algorithms (A1–A5)

- A1: Write recursive and non-recursive programs to calculate Fibonacci numbers and analyze time and space complexity.
- A2: Implement Huffman Encoding using a greedy strategy.
- A3: Solve the 0-1 Knapsack problem using dynamic programming or branch and bound.
- A4: Design an n-Queens matrix with the first queen already placed; use backtracking to place the remaining queens.
- A5: Analyze Quick Sort using deterministic and randomized variants.

Group B: Machine Learning (B1–B5)

- B1: Predict Uber ride fare: preprocess data, identify outliers, check correlation, train Linear Regression and Random Forest Regression, and compare R^2 and RMSE.
- B2: Classify email as spam or not spam using K-Nearest Neighbors and Support Vector Machine; analyze performance.
- B3: Build a neural-network classifier to predict bank customer churn within six months; split and normalize data, train the model, and print accuracy and confusion matrix.
- B4: Implement Gradient Descent to find a local minimum of $y = (x + 3)^2$ starting at $x = 2$.
- B5: Apply K-Means or hierarchical clustering to sales_data_sample.csv; determine the number of clusters using the elbow method.

Group C: Blockchain Technology (C1–C2)

- C1: Write and deploy a bank account smart contract on a test network supporting deposit, withdrawal, and balance display.
- C2: Write and deploy a Solidity student-data smart contract using structures and arrays; observe transaction fee and gas values.

4. Blockchain Technology: Setup and Use

Recommended development environment: Remix IDE Desktop. Download and install it before the examination from https://remix-ide.readthedocs.io/en/latest/desktop.html while practicing. Ensure that Solidity compiler version 0.8.38 is available.

The listed C1 and C2 assignments specify test-network deployment and/or observing transaction fee and gas values. Completing those requirements on an external test network requires internet connectivity and the appropriate network and wallet configuration.

5. Preparation Checklist

- Practise all relevant problem statements from the groups that may appear in your assignment combination.
- Check the programming environment and required language/compiler versions during practice.

- For Machine Learning, practise loading datasets, running preprocessing steps, training the required models, and reporting the requested evaluation results.
- Ensure that required datasets, Python packages, and NLTK corpora are available in the environment used for practice.
- For Blockchain Technology, practise compiling contracts, deploying them in the permitted environment, calling contract functions, and checking gas/transaction information as applicable.
- Read the assigned problem statement carefully during the examination and confirm any environment-related instructions with the examiner before starting.

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